



Technical Service Bulletin

Technical Service Bulletin: TSB210067

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Revised Engine Cylinder Blocks

Revised Engine Cylinder Blocks

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

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- QSL9 G CM558 L104

Description of Change

- This document announces the introduction of revised cylinder blocks.
- The changes that impact service include cylinder head mounting capscrew counterbore depth and associated cylinder head capscrew part supersession.

Service Instructions

- New cylinder blocks and their corresponding cylinder head mounting capscrews are **not** interchangeable with existing cylinder blocks and their associated cylinder head mounting capscrews.
- Torque procedure will vary dependent upon the combination of cylinder head capscrews and cylinder head type.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Cylinder Block	5271268	Yes	Yes	5619266
Cylinder Block	5293397	Yes	Yes	5619268
Cylinder Block	5524176	Yes	Yes	5619270
Cylinder Block	5289666	Yes	Yes	5619272
Cylinder Block	5293399	Yes	Yes	5619274
Cylinder Block	5293401	Yes	Yes	5619276
Cylinder Block	5304906	Yes	Yes	5619278
Cylinder Block	3971411	Yes	Yes	5619280
Cylinder Block	-	-	-	5561276
Cylinder Head Capscrew	3960043	No	Yes	5482243
Cylinder Head Capscrew	3917729	No	Yes	5561753

Part Compatibility

- Cylinder head mounting capscrew selection will require special attention when replacing parts.
- Option numbers and service kit part numbers will **not** change. **Only** the part structures within will change.
- Cylinder head capscrew replacement will be driven by which cylinder block is being used. Cylinder head part number is independent of cylinder block/cylinder head capscrew

compatibility.

- The cylinder head capscrew torque procedure will vary dependent upon which type of cylinder head and cylinder head capscrews are being installed. See Table 2.

Table 2, Torque Specifications

Cylinder Head Type	Capscrew Part Number	Final Torque Procedure
4-Valve Cylinder Head	3960043	115 N•m [85 ft-lb] + 120 degrees
	5482243	115 N•m [85 ft-lb] + 150 degrees
2-Valve Cylinder Head	3960043	142 N•m [105 ft-lb] + 90 degrees
	3917729	70 N•m [52 ft-lb] + 90 degrees
	5482243	165 N•m [122 ft-lb] + 90 degrees
	5561753	80 N•m [59 ft-lb] + 90 degrees

Part Identification

New cylinder head capscrews can be identified by a white paint marking on the top surface before first use.

Part Inventory Action

Existing cylinder blocks should use existing cylinder head capscrew part numbers. New cylinder blocks should use new cylinder head capscrew part numbers. See Table 2 above.

Production Status

Implemented for production. See Table 3.

Table 3, Production Information

ESN First	Build Date ¹	Plant
74705894	28 September 2020	Rocky Mount Engine Plant

¹Engine build date can be found on the engine dataplate.

Publications Affected

Table 4, Publications Affected

Manual Type	Bulletin Number	Procedure Title	Procedure
QSL9 G CM558 Service Manual	2883375	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/147/147-002-004-tr- qsl9g.html)

Table 4, Publications Affected			
Manual Type	Bulletin Number	Procedure Title	Procedure
C Series Troubleshooting and Repair Manual	3666003	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html)
C8.3G, C Gas Plus, and L Gas Plus Troubleshooting and Repair Manual	3666206	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html)
ISC, ISCe, QSC8.3, ISL, ISLe3, ISLe4 and QSL9 Troubleshooting and Repair Manual	4021418	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr.html)
ISC and ISL CM2150 Service Manual	4021569	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-iscl07.html)
ISLe CM2150 Service Manual	4021630	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-isclw.html)
ISL G CM2180 Service Manual	4021649	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-islg.html)
ISL9 CM2150 SN Service Manual	4022190	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-isle5.html)

Table 4, Publications Affected			
Manual Type	Bulletin Number	Procedure Title	Procedure
QSL9 CM2250 Service Manual	4022256	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-qlt4i.html)
ISC8.3 and ISL9 CM2250 Service Manual	4022257	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/45/45-002-004-tr-iscl10.html)
L8.9 Series Troubleshooting and Repair Manual	4310704	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/183/183-002-004.html)
ISL9 CM2350 L101 Service Manual	4310787	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/185/185-002-004.html)
QSL9 G CM558 L104 Service Manual	4326000	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/229/229-002-004-tr.html)
QSL9 CM2350 L102 Service Manual	4332796	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/261/261-002-004-tr.html)
QSL9 M CM2250 L106 Service Manual	4358343	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/263/263-002-004-tr.html)

Table 4, Publications Affected			
Manual Type	Bulletin Number	Procedure Title	Procedure
ISL8.9 CM2150 L110 Service Manual	4358475	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/323/323-002-004-tr.html)
ISL8.9 CM2880 L112 Service Manual	4358493	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/336/336-002-004-tr.html)
ISL9 CM2350 L111 Service Manual	4367223	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/341/341-002-004-tr.html)
ISC8.3 G CM2180 C101 Service Manual	4367326	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/361/361-002-004.html)
QSL9 CM2350 L107 Service Manual	4367386	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/333/333-002-004-tr.html)
QSL9 CM2250 L115 Service Manual	4388626	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/409/409-002-004.html)
QSL9 CM2350 L118 Service Manual	4388673	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/421/421-002-004.html)
QSC8.3 CM2880 C102 Service Manual	4388785	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/436/436-002-004.html)

Table 4, Publications Affected			
Manual Type	Bulletin Number	Procedure Title	Procedure
L8.9 L121 Service Manual	5411055	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/447/447-002-004.html)
L9 CM2350 L119B Service Manual	5411267	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/425/425-002-004.html)
QSL9 CM2350 L122 Service Manual	5411278	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/487/487-002-004.html)
L9 CM2350 L116B Service Manual	5411330	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/459/459-002-004.html)
L9 CM2350 L123B Transit Bus Applications Service Manual	5411465	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/459/459-002-004.html)
L9N CM2380 L124B and L9N CM2380 L130B Service Manual	5467492	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/521/521-002-004.html)
L9 CM2350 L120C Service Manual	5504199	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/560/560-002-004.html)
L9 CM2670 L128B Service Manual	5504589	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/622/622-002-004.html)

Table 4, Publications Affected			
Manual Type	Bulletin Number	Procedure Title	Procedure
L9 CM2350 L129C Service Manual	5579689	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/639/639-002-004.html)
L9 CM2350 L133C Service Manual	5613155	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/727/727-002-004.html)
L9 CM2450 L126B Service Manual	5613077	Cylinder Head	Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/583/583-002-004.html)

Document History

Date	Details
2021-3-26	Module Created

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